

RAJESH KUMAR DASH

Lead Data Engineer | Databricks • Spark • Yarn • HDFS • Cloudera • Cloud (AWS & Azure)

Bangalore, 560067, India | Available for Remote/Hybrid Roles

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PROFESSIONAL SUMMARY

Lead Data Engineer with 12 years of experience designing enterprise-grade data platforms distributed data systems and strong problem-solving skills across domains including **Semiconductor, Telecom, Automotive, Healthcare**. Proven track record of leading architecture decisions, defining data engineering standards, and delivering highly reliable Spark and Databricks-based platforms on AWS and Azure.

CORE DATA ENGINEERING & LEADERSHIP SKILLS

- **Platforms:** Databricks (AWS & Azure), Delta Lake, Unity Catalog, Snowflake, Cloudera
- **Architecture:** Medallion / Lakehouse, Dimensional Modeling, SCD Type-2
- **Data Engineering:** Advanced SQL, Batch & Incremental ETL/ELT, CDC, Schema Evolution, Query performance tuning, join strategies, and cost-aware data modeling, Hadoop, Cloudera, Hive, Pandas, Numpy, Apache Spark
- **Quality & Governance:** Data Quality Frameworks (Z-score, statistical checks), Lineage
- **IDE:** Vs-Code, IntelliJ-Idea, Copilot, Cursor, Claude Code
- **Streaming:** Kafka, Kinesis Firehose, SQS, SNS
- **Databases:** PostgreSQL, MS SQL, Cassandra, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS, Azure, Terraform, Jenkins, GitHub Actions, Docker, Kubernetes
- **Agentic AI:** Crew AI, Lang Chain
- **Languages:** Scala, Python, Java, TypeScript

PROFESSIONAL EXPERIENCE

Technical Lead – Laerdal Medical (2021 – current)

- Architected and owned **production-grade Databricks Lakehouse** using **Spark, Delta Lake, and Unity Catalog** across AWS and Azure, supporting analytics and AI/ML workloads.
- Handled schema evolution using data contracts, enabling backward-compatible processing and re-parsing of historical data.
- Automated support channel requests through an Azure Bot that creates a support ticket and notifies the designated support roster member for the week.

- Developed a POC for synthetic data generation using agentic ai (crew ai) for the lower environment by preserving data distributions from the data warehouse and replicating SCD Type 2 (SCD2) patterns.
- Designed **reusable ingestion frameworks** for APIs, RDBMS, and file-based sources (JSON, XML, CSV), reducing onboarding time for new datasets.
- Led **governance and compliance implementation** including **GDPR and FedRAMP**, enabling secure PII handling through anonymization and column-level access control.
- Built a **configurable data quality framework** with **statistical validations (Z-score, threshold checks)**, improving data reliability for downstream analytics and ML.
- Implemented **SCD Type-2 dimensional models** to support historical analytics and audit requirements.
- Migrated **small-file batch workloads** to **AWS Lambda + Kinesis Firehose (Python, Typescript)**, significantly reducing compute cost and end-to-end latency.
- Automated Databricks workspaces, jobs, and AWS infrastructure using **Terraform**, standardizing environments across teams.
- **Led AWS → Azure Databricks migration**, ensuring zero data loss and minimal downtime.
- Acted as **technical mentor and design reviewer** for junior engineers, setting data engineering standards while remaining hands-on.

AI / Analytics Enablement

- Designed **analytics- and ML-ready curated datasets** by enforcing schema consistency, quality gates, and governance controls.
- Enabled **trusted feature consumption** by downstream data science and analytics teams through clean Delta tables and versioned data.
- Contributed to **AI readiness** by ensuring data compliance, lineage, and reproducibility across environments.

Module Lead – American Express (01/2021-06/2021)

- Built **Spark-based ETL pipelines** for **enterprise security analytics (CISO)**.
- Integrated **NiFi, Elasticsearch, and CrowdStrike** for threat detection workflows from kafka based system.
- Maintaining Hive based Datawarehouse for Historical Data analysis.

Senior Engineer – HCL Software / Sametime (05/2018 –11/ 2020)

- Led backend development for **Sametime enterprise chat application**, building and maintaining **Java-based microservices** supporting real-time messaging at scale and developed **Chat History Service**, responsible for message persistence, retrieval using **Mongo-Db**
- Designed and implemented a reward-loader service to process high-volume telecom recharge events and calculate customer rewards using Hadoop and Hive.

Software Engineer – NXP Semiconductor (07/2013 –02/2018)

Designed and developed Java-based Eclipse RCP internal tools for semiconductor chip testing, along with a PoC for manufacturing data collection to improve defect detection and quality assurance using Kafka and Hadoop.

EDUCATION

MS Data Science(Global) – Deakin University, Australia (2025)

PG Certification in AI & ML – University of Texas at Austin (2025)

Bachelor of Technology, Computer Science – BPUT (2012)